問題 326

次の値を求めよ．

解答:

(1) $\cos\dfrac{5\pi}{12} = \cos(75^\circ) = \cos(45^\circ + 30^\circ) = \dfrac{\sqrt{6}-\sqrt{2}}{4}$

(2) $\sin\dfrac{\pi}{12} = \sin(15^\circ) = \sin(45^\circ - 30^\circ) = \dfrac{\sqrt{6}-\sqrt{2}}{4}$

(3) $\tan\dfrac{7\pi}{12} = \tan(105^\circ) = \tan(60^\circ + 45^\circ) = \dfrac{\sqrt{3}+1}{1-\sqrt{3}} = -(2+\sqrt{3})$

問題 327

$\sin\alpha = \dfrac{3}{5}$（$\alpha$ 鋭角），$\cos\beta = -\dfrac{12}{13}$（$\beta$ 鈍角）のとき，$\cos(\alpha+\beta)$ を求めよ．

解答:

$\cos\alpha = \dfrac{4}{5}$，$\sin\beta = \dfrac{5}{13}$

$\cos(\alpha+\beta) = \dfrac{4}{5}\cdot\left(-\dfrac{12}{13}\right) - \dfrac{3}{5}\cdot\dfrac{5}{13} = -\dfrac{48}{65} - \dfrac{15}{65} = -\dfrac{63}{65}$

問題 328

θ が第 1 象限で $\sin\theta = \dfrac{5}{13}$ のとき，$\sin 2\theta$，$\cos 2\theta$ を求めよ．

解答:

$\cos\theta = \dfrac{12}{13}$

$\sin 2\theta = 2\cdot\dfrac{5}{13}\cdot\dfrac{12}{13} = \dfrac{120}{169}$

$\cos 2\theta = 1 - 2\cdot\dfrac{25}{169} = \dfrac{119}{169}$

問題 329

$0 \leqq \theta < 2\pi$ で $\sin\theta + \sqrt{3}\cos\theta = -1$ を解け．

解答:

$\sin\theta + \sqrt{3}\cos\theta = 2\sin\left(\theta + \dfrac{\pi}{3}\right) = -1$

$\sin\left(\theta + \dfrac{\pi}{3}\right) = -\dfrac{1}{2}$

$\theta + \dfrac{\pi}{3} = \dfrac{7\pi}{6},\ \dfrac{11\pi}{6}$

$\theta = \dfrac{5\pi}{6},\ \dfrac{3\pi}{2}$

問題 330

$0 \leqq \theta < 2\pi$ のとき，$y = \sin\theta - \sqrt{3}\cos\theta$ の最大値と最小値を求めよ．

解答:

$y = 2\sin\left(\theta - \dfrac{\pi}{3}\right)$

最大値 2 ($\theta - \dfrac{\pi}{3} = \dfrac{\pi}{2}$, $\theta = \dfrac{5\pi}{6}$)，最小値 -2 ($\theta = \dfrac{11\pi}{6}$)

問題 331

$\tan\alpha + \tan\beta = \dfrac{\sin(\alpha+\beta)}{\cos\alpha\cos\beta}$ を証明せよ．

解答:

左辺 $= \dfrac{\sin\alpha}{\cos\alpha} + \dfrac{\sin\beta}{\cos\beta} = \dfrac{\sin\alpha\cos\beta + \cos\alpha\sin\beta}{\cos\alpha\cos\beta} = \dfrac{\sin(\alpha+\beta)}{\cos\alpha\cos\beta} =$ 右辺 □

問題 332

$\sin(\alpha+\beta)\sin(\alpha-\beta) = \sin^2\alpha - \sin^2\beta$ を証明せよ．

解答:

左辺 $= (\sin\alpha\cos\beta + \cos\alpha\sin\beta)(\sin\alpha\cos\beta - \cos\alpha\sin\beta)$

$= \sin^2\alpha\cos^2\beta - \cos^2\alpha\sin^2\beta$

$= \sin^2\alpha(1-\sin^2\beta) - (1-\sin^2\alpha)\sin^2\beta$

$= \sin^2\alpha - \sin^2\alpha\sin^2\beta - \sin^2\beta + \sin^2\alpha\sin^2\beta = \sin^2\alpha - \sin^2\beta =$ 右辺 □

問題 333

2 直線 $y = \dfrac{1}{2}x + 1$，$y = 2x - 3$ のなす角 θ を求めよ．

解答:

傾き $m_1 = \dfrac{1}{2}$，$m_2 = 2$ とすると

$\tan\theta = \left|\dfrac{m_1 - m_2}{1 + m_1 m_2}\right| = \left|\dfrac{\frac{1}{2} - 2}{1+1}\right| = \left|\dfrac{-\frac{3}{2}}{2}\right| = \dfrac{3}{4}$

$\theta = \arctan\dfrac{3}{4} \approx 36.87^\circ$

問題 334

$\sin 2\theta = 2\sin\theta\cos\theta$ と $\cos 2\theta = \cos^2\theta - \sin^2\theta$ を用いて $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$ を導け．

解答:

$\tan 2\theta = \dfrac{\sin 2\theta}{\cos 2\theta} = \dfrac{2\sin\theta\cos\theta}{\cos^2\theta - \sin^2\theta}$

分子分母を $\cos^2\theta$ で割る：

$= \dfrac{2\frac{\sin\theta}{\cos\theta}}{1 - \frac{\sin^2\theta}{\cos^2\theta}} = \dfrac{2\tan\theta}{1-\tan^2\theta}$ □

問題 335

$\cos^4\theta - \sin^4\theta = \cos 2\theta$ を示せ．

解答:

左辺 $= (\cos^2\theta + \sin^2\theta)(\cos^2\theta - \sin^2\theta) = 1\cdot\cos 2\theta =$ 右辺 □

問題 336

$0 \leqq \theta < 2\pi$ のとき $2\cos^2\theta + \sin\theta - 1 = 0$ を解け.

解答:

$\cos^2\theta = 1 - \sin^2\theta$ を代入：

$2(1-\sin^2\theta) + \sin\theta - 1 = 0,\ -2\sin^2\theta + \sin\theta + 1 = 0$

$2\sin^2\theta - \sin\theta - 1 = 0,\ (2\sin\theta + 1)(\sin\theta - 1) = 0$

$\sin\theta = -\dfrac{1}{2}$ または $\sin\theta = 1$

$\sin\theta = 1$：$\theta = \dfrac{\pi}{2}$

$\sin\theta = -\dfrac{1}{2}$：$\theta = \dfrac{7\pi}{6},\ \dfrac{11\pi}{6}$

$\therefore \theta = \dfrac{\pi}{2},\ \dfrac{7\pi}{6},\ \dfrac{11\pi}{6}$